# Exercise 1

Given exchangeable data set $(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$, level $\alpha \in (0, 1)$, and a real valued symmetric score function $s((x, y); \mathcal{D})$ for $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and $\mathcal{D} \in (\mathcal{X} \times \mathcal{Y})^k$ for any $k$ being a data set, we carry out the full conformal prediction. Put, for any $y \in \mathcal{Y}$:

$$\mathcal{D}_{n+1}^y = \{(X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, y)\}.$$

For simplicity, write $S_i^y = s((X_i, Y_i); \mathcal{D}_{n+1}^y)$ for $i \in \{1, \ldots, n\}$ and $S_{n+1}^y = s((X_{n+1}, y); \mathcal{D}_{n+1}^y)$. The standard conformal prediction procedure asks us to compare $S_{n+1}^y$ with the entire sequence $(S_1^y, \ldots, S_n^y)$.

We proceed however by choosing randomly a subset $I \subseteq \{1, \ldots, n\}$ independent of the data and compare $S_{n+1}^y$ with $(S_i^y)_{i \in I}$. More precisely, put:

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} \mid S_{n+1}^y \leqslant \text{Quantile}\left( (S_i^y)_{i \in I}; (1-\alpha)\left(1 + \frac{1}{|I|}\right)\right)\right\}.$$

Prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

**Solution** Given that $I$ is independent of the data set, the joint distribution of $((X_1, Y_1), \cdots, (X_n, Y_n), (X_{n+1}, Y_{n+1}))$ remains the same after conditioning on $I$. In particular, they are still exchangeable. Therefore:

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} = \sum P\{I = J\} P\left(\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} \mid I = J\right) \geqslant \sum P\{I = J\}(1-\alpha) = 1 - \alpha,$$

where the summation is taken with $J$ ranged over all subsets of $1, \ldots, n$. The second inequality is Theorem 3.2 in the textbook. ////